CASE 06 · ADVANCED BUSINESS ANALYST PROJECT · BANKING

The 182% Channel Gap

A statistical hypothesis-testing case study on a real bank telemarketing campaign — plus the A/B test it should have been.

Role simulated: Senior Business Analyst / Marketing Analytics | Tools: Python (scipy, statsmodels — chi-square, z-test, power analysis), Excel

Source: UCI Bank Marketing dataset (public) — 41,188 real telemarketing contacts, Portuguese retail bank

14.7%	5.2%	+182%	0.326
Cellular-channel conversion rate	Telephone-channel conversion rate	Relative lift, cellular vs. telephone	Effect size (Cohen's h)

The business problem

A retail bank's marketing team wants to know whether contact channel meaningfully affects term-deposit conversion, to decide where to invest calling capacity next quarter. "Cellular looks better in the numbers" isn't good enough to move budget on — this project puts an actual statistical test behind it, and is honest about what that test can and can't prove.

The test

Of 26,144 contacts made by cellular, 3,853 converted (14.74%, 95% CI [14.3%, 15.2%]). Of 15,044 contacts made by telephone, 787 converted (5.23%, 95% CI [4.9%, 5.6%]). A chi-square test of independence (chi-square statistic = 862.3, $p < 0.001$) and a two-proportion z-test ($z = 29.4$, $p < 0.001$) both reject the null with essentially no room for doubt — at this sample size, the confidence intervals don't come close to overlapping.

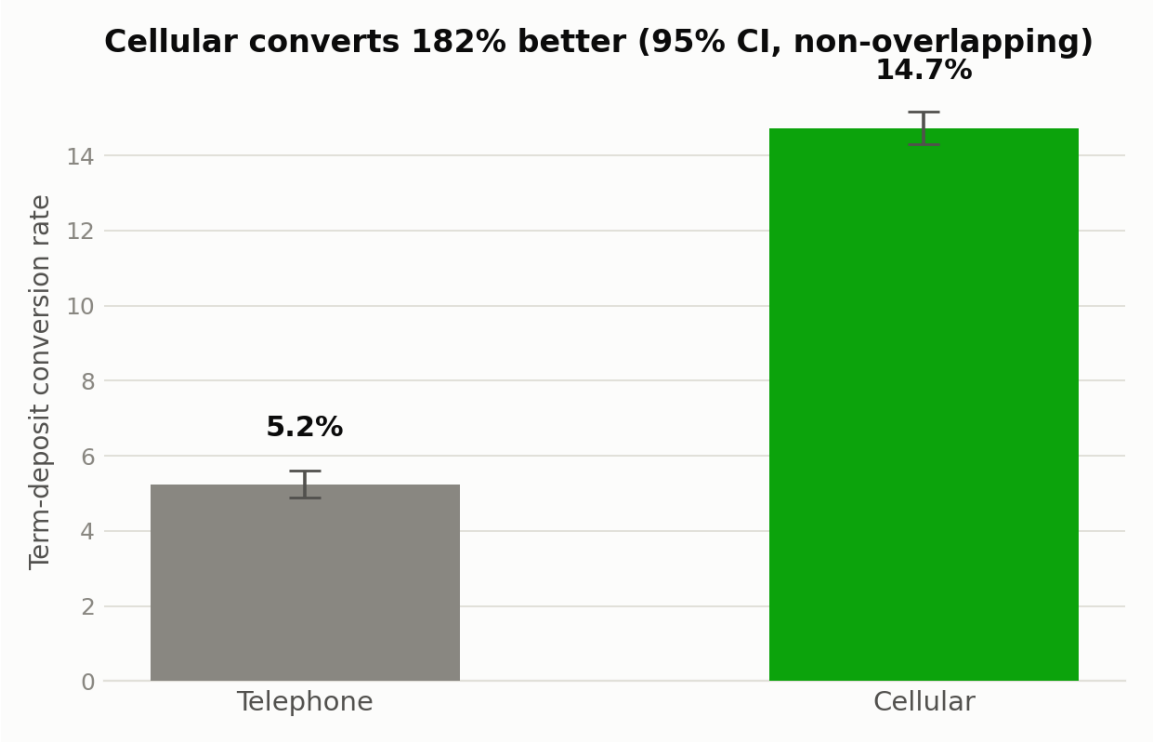


Fig. 1 — Conversion rate by channel with 95% confidence intervals.

The catch — and why it matters more than the p-value

A p-value this small is easy to over-read as "proof cellular works better." It isn't, and saying so plainly is the actual senior-analyst move here: contact channel in this dataset was **not randomly assigned** — it's a retrospective record of which channel field agents happened to use, contact by contact, likely correlated with things like campaign timing, target list quality, and which customers even own a cell phone. The statistical test confirms the gap is real and not noise; it says nothing about whether switching a telephone-contacted customer to cellular would produce the same lift. That requires an actual experiment.

One supporting data point worth flagging on the confounding question: prior-campaign outcome is an even stronger predictor than channel — customers with a successful prior campaign convert at 65.1% vs. 9.4% otherwise ($p < 0.001$). If channel assignment correlates at all with who's been successfully contacted before, that alone could explain part of the channel gap.

Designing the real test

The recommendation isn't "trust this number" — it's "randomize channel assignment for one quarter and re-run this exact test." The accompanying Excel calculator turns the observed baseline into a sample-size plan for that future test: bigger minimum-detectable effects need dramatically fewer contacts per arm.

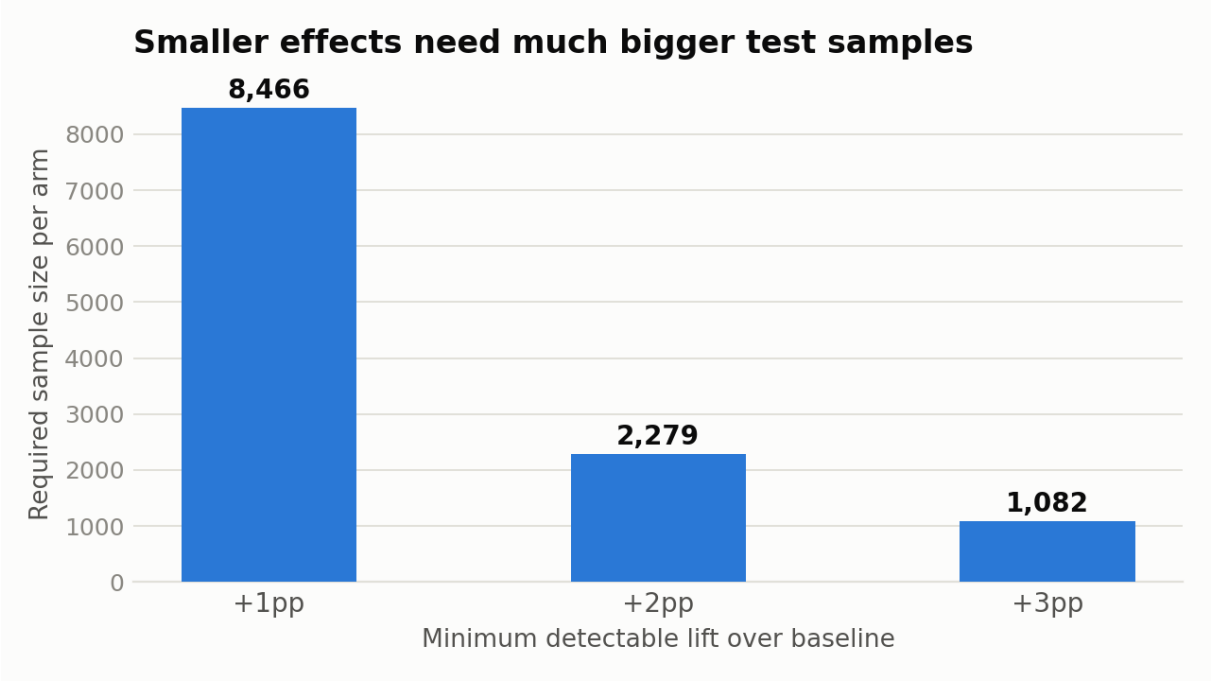


Fig. 2 — Required sample size per arm to detect a given lift at 95% confidence / 80% power.

Minimum detectable lift	Sample size needed per arm	Total contacts
+1 percentage point	~8,466	~16,932
+2 percentage points	~2,286	~4,572
+3 percentage points	~1,090	~2,179

From case6_ab_test_calculator.xlsx — live formulas, edit the baseline or MDE and every row recalculates.

Recommendation

Don't reallocate the calling budget off the observational gap alone, however tempting the p-value looks. Run a randomized pilot — even the smallest, ~1,090-per-arm design (detecting a 3-point lift) is well within a normal quarter's contact volume — and let a true experiment settle whether channel is causal or just correlated with something else already happening in the campaign.

RESUME BULLET, READY TO USE

“Ran chi-square and two-proportion z-tests on 41,188 real bank marketing contacts, finding a statistically significant 182% conversion gap between contact channels ($p < 0.001$); identified the result as observational and designed a randomized follow-up test with a live Excel sample-size calculator.”

What's in this package

File	What it is
run_analysis.py	Chi-square test, z-test, effect size, and power analysis — fully reproducible.
case6_ab_test_calculator.xlsx	Live sample-size calculator for the proposed randomized follow-up test.
This report (PDF)	Test results, the confounding caveat, and the experiment design.